Kai Zheng

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EDUCATION BACKGROUND

University of Southern California *08/2023-12/2025*

Master of Science in Computer Science

- **GPA: 3.92**
- **Highlighted Courses:** Web Technology, Multimedia System Design, Database System, Advanced Data Store, Analysis of Algorithms, Deep Learning, Advanced Computer Vision

University of California, Los Angeles *Graduated in 06/2023*

Bachelor of Science in Mathematics of Computation, Minor in Statistics and Data Science

- **GPA: 3.76**
- **Highlighted Courses:** Software Construction, Introduction to Algorithms and Complexity, Operation Systems Principles, Programming Languages, Computer Architecture

INTERNSHIP EXPERIENCE

Circue Energy *06/2024-08/2024*

Software Engineer Intern

- Developed algorithm tools for energy storage system monitoring to provide timely warnings for abnormal situations such as thermal runaway.
- Conducted interpretability research on the vehicle battery anomaly detection model by leveraging the characteristics of the SVM algorithm and recording gradient backpropagation, allowing for a detailed understanding of key data features impacting the model. Visualized and documented this information to help users better comprehend data characteristics.
- Responsible for updating database labels.

Institute of Computing Technology of Chinese Academy of Sciences *07/2022-08/2022*

Software Engineer Intern

- Reviewed and summarized research papers on detecting variable associations in large datasets, assisting the advisor in organizing various statistical methods and their applications in real-world datasets.
- Implemented nine traditional correlation measurement algorithms using Python, including Cor, Scor, Dcor, HHG, MIC, GR, RDC, XIco, and ACE, as well as two novel correlation measurement algorithms: Cauchy Combination and Harmonic Mean P-value.
- Applied these methods to econometric and public health data analysis, improving the efficiency of correlation testing.

PROJECT EXPERIENCE

Stock Account Website and Mobile Application Development *02/2024-05/2024*

- Designed and developed a website allowing users to search for stock information and simulate stock market trading, along with an iOS version with the same functionality.
- Utilized TypeScript, HTML, CSS, and the Angular framework to design the front-end web pages.
- Designed the back-end using Express.js, and created a RESTful API for front-end interactions.
- Employed MongoDB to provide and store stock data for the front-end.
- Developed the iOS application using Swift for iPhone users.

C++ High-Concurrency Cache System *10/2023-02/2024*

- Implemented **multiple cache replacement policies** to adapt to different access patterns and business scenarios.
- Developed **cache sharding for LRU and LFU** to reduce lock contention and improve performance under high-concurrency access.
- Enhanced **LRU-k optimization** to prevent hot data from being replaced by cold data, reducing cache pollution.
- Introduced **maximum average access frequency for LFU** to evict outdated hot data, improving overall cache utilization.
- Implemented **ARC strategy** to dynamically adjust the weight ratio of LRU and LFU, improving cache hit rates in complex environments.
- Achieved **thread safety** through the use of mutex locks and atomic operations for multi-threaded access.

GRIDS Mindspark Data Science Hackathon, *Won Second Place* *09/2023*

*Team Member: **Kai Zheng,** Zijie Lei, Huixian Gong, Zhenyi Xie,*

- Designed a CNN-based model to detect potholes on the road, achieving a 90 percent accuracy rate
- Designed data processing pipline that collected and cleaned more than 1000 pictures of road captured by the Driving Recorder
- Build the CNN model using TensorFlow
- Designed exploratory data analysis using KNIME

Paper in Computer Vision

*Sarkar, S., Kundu, S., **Zheng, K.**, Beerel, P. 2024.* ***Block Selective Reprogramming for On-device Training of Vision Transformers***

CVPR 2024 Efficient Computer Vision

We propose **Block Selective Reprogramming**, a method that fine-tunes a fraction of model blocks and selectively drops tokens, achieving up to 1.4x reduction in training memory and 2x reduction in computation cost while maintaining accuracy across diverse datasets and multitask learning scenarios.

ADDITIONAL INFORMATION

Programming Skills: C/C++ , Python (Flask), R, JavaScript, TypeScript, Node.js,

Tools: PostgreSQL, MySQL, MongoDB, Node.js, Express.js, Google Cloud Platform, Angular, React, Docker, Unix/Linux, Github, Flask

Languages: English (proficient, score "A" in Academic Writing), Chinese Mandarin (native)